



3017 - High-resolution imaging of a compact lensed quasar at $z=5.07$ and a compound lensing system

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) J0025-0145

ABSTRACT

High-redshift lensed quasars are powerful tools to probe the evolution of SMBHs and their host galaxies in the early universe. The population of lensed quasars at high redshifts is largely unexplored, with only one lensed quasar reported at $z>5$ (J0439+1634 at $z=6.52$). Recently, J0025-0145 ($z=5.07$) has been identified as a lensed quasar, which is the second lensed quasar discovered at $z>5$. HST imaging unambiguously shows the existence of the foreground lensing galaxy, and the high apparent Eddington ratio suggests that J0025-0145 has a high magnification ($>\sim 80$). However, the lensing structure of J0025-0145 is unresolved even in the HST images; instead of multiple lensed images, the quasar appears to be a single extended object. Here we propose NIRCam imaging for J0025-0145 in the F070W and the F480M filters. The F070W imaging delivers a PSF size that is $\sim 3\times$ sharper than HST and will safely resolve the lensing structure; the F480M filter detects long wavelengths where the quasar host

galaxy has prominent flux. We will construct an accurate lensing model and measure the magnification for the background quasar. Aided by lensing magnification, J0025-0145 provides an exceptional chance to measure the stellar emission and morphology of a high-redshift quasar host galaxy, shedding light on the SMBH-host coevolution in the early universe. The proposed observation will yield a bonus science, i.e., constraining the properties of dark matter and dark energy using a triple-source compound lensing system in the field. With a very modest time request of 2.49 hours, this proposal will lead to high-impact results in both high-redshift quasar studies and cosmology.

OBSERVING DESCRIPTION

We request NIRCам imaging of a lensed quasar at $z=5.07$ to resolve its lensing structure and detect its host galaxy emission. We will observe with F070W filter in the SW channel and the F480M filter in the LW channel:

- The F070W filter delivers the sharpest PSF that JWST can provide, which guarantees that the lensing structure will get resolved;
- The F480M filter detects the wavelengths where the quasar host galaxy has prominent flux. A medium filter is used to avoid saturation.

We will apply the INTRAMODULEBOX primary dither and the STANDARD 9-point dither. We will read the FULL array with the RAPID readout pattern, setting $N_{\text{group}}=5$ and $N_{\text{int}}=1$ for each dither position. The observation only contains one pointing and can be executed in a single visit.

Proposal 3017 - Targets - High-resolution imaging of a compact lensed quasar at $z=5.07$ and a compound lensing system

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	J0025-0145	RA: 00 25 27.2032 (6.3633467d) Dec: -01 45 2.57 (-1.75071d) Equinox: J2000		
Comments: Category=Galaxy Description=[Quasars]					

Proposal 3017 - Observation 1 - High-resolution imaging of a compact lensed quasar at z=5.07 and a compound lensing system

Observation	Proposal 3017, Observation 1									Sat May 27 01:00:31 GMT 2023
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(1)	J0025-0145	RA: 00 25 27.2032 (6.3633467d)							
			Dec: -01 45 2.57 (-1.75071d)							
	Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars]									
Template	Module		Subarray				Target Placement			
	B		FULL				Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX		4		STANDARD				9
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	F070W	F480M	RAPID	5	1	36	36	1932.619	